

| Question | Marking Guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Mark  | Comments                                                                                                                                                                                                                                                                    |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 09.1     | <p><b>Comparisons</b></p> <ol style="list-style-type: none"> <li>1. Nucleotide structure is identical;</li> <li>2. Nucleotides joined by phosphodiester bond;</li> </ol> <p><b>OR</b></p> <p>Deoxyribose joined to phosphate (in sugar, phosphate backbone);</p> <ol style="list-style-type: none"> <li>3. DNA in mitochondria / chloroplasts same / similar (structure) to DNA in prokaryotes;</li> </ol> <p><b>Contrasts</b></p> <ol style="list-style-type: none"> <li>4. Eukaryotic DNA is longer;</li> <li>5. Eukaryotic DNA contain introns, prokaryotic DNA does not;</li> <li>6. Eukaryotic DNA is linear, prokaryotic DNA is circular;</li> <li>7. Eukaryotic DNA is associated with / bound to protein / histones, prokaryotic DNA is not;</li> </ol> | 5 max | <ol style="list-style-type: none"> <li>1. Accept labelled diagram or description of nucleotide as phosphate, deoxyribose and base</li> <li>3. Accept shorter than nuclear DNA/is circular not linear/is not associated with protein/histones unlike nuclear DNA;</li> </ol> |
| 09.2     | <ol style="list-style-type: none"> <li>1. Mutations change base / nucleotide (sequence);</li> <li>2. (Causing) change in amino acid sequence;</li> <li>3. Mutations build up over time;</li> <li>4. More mutations / more differences (in amino acid / base / nucleotide sequence / primary structure) between distantly related species;</li> </ol> <p><b>OR</b></p> <p>Few(er) mutations / differences (in amino acid / base / nucleotide sequence / primary structure) in closely related species;</p> <ol style="list-style-type: none"> <li>5. Distantly related species have earlier common ancestor;</li> </ol> <p><b>OR</b></p> <p>Closely related species have recent common ancestor;</p>                                                             | 5     | <p>Accept "order" for "sequence"</p> <ol style="list-style-type: none"> <li>1. Reject if mutation in amino acid</li> </ol> <p>If neither MP4 or MP5 accept for 1 mark, idea of more mutations /differences as evidence of earlier common ancestor OR converse</p>           |